

BCSE309L	Cryptography and Network Security	L	T	P	C
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Pre-requisite	NIL	Syllabus version			
		1.0			
Course Objectives					
1. To explore the concepts of basic number theory and cryptographic techniques.					
2. To impart concept of Hash and Message Authentication, Digital Signatures and authentication protocols.					
3. To reveal the basics of transport layer security, Web Security and various types of System Security.					
Course Outcomes					
On completion of this course, students should be able to:					
1. To know the fundamental mathematical concepts related to security.					
2. To understand concept of various cryptographic techniques.					
3. To apprehend the authentication and integrity process of data for various applications					
4. To know fundamentals of Transport layer security, web security, E-Mail Security and IP Security					
Module:1 Fundamentals of Number Theory		5 hours			
Finite Fields and Number Theory: Modular arithmetic, Euclidian Algorithm, Primality Testing: Fermats and Eulers theorem, Chinese Remainder theorem, Discrete Logarithms.					
Module:2 Symmetric Encryption Algorithms		7 hours			
Symmetric key cryptographic techniques: Introduction to Stream cipher, Block cipher: DES, AES,IDEA, Block Cipher Operation, Random Bit Generation and RC4					
Module:3 Asymmetric Encryption Algorithm and Key Exchange		8 hours			
Asymmetric key cryptographic techniques: principles, RSA, ElGamal, Elliptic Curve cryptography, Homomorphic Encryption and Secret Sharing, Key distribution and Key exchange protocols, Diffie-Hellman Key Exchange, Man-in-the-Meddle Attack					
Module:4 Message Digest and Hash Functions		5 hours			
Requirements for Hash Functions, Security of Hash Functions, Message Digest (MD5), Secure Hash Function (SHA), Birthday Attack, HMAC					
Module:5 Digital Signature and Authentication Protocols		7 hours			
Authentication Requirements, Authentication Functions, Message Authentication Codes, Digital Signature Authentication, Authentication Protocols, Digital Signature Standards, RSA Digital Signature, Elgamal based Digital Signature, Authentication Applications: Kerberos, X.509 Authentication Service, Public Key Infrastructure (PKI)					
Module:6 Transport Layer Security and IP Security		4 hours			
Transport-Layer Security, Secure Socket Layer(SSL), TLS, IP Security: Overview: IP Security Architecture, Encapsulating Payload Security					
Module:7 E-mail, Web and System Security		7 hours			
Electronic Mail Security, Pretty Good Privacy (PGP), S/MIME, Web Security: Web Security Considerations, Secure Electronic Transaction Protocol Intruders, Intrusion Detection, Password Management, Firewalls: Firewall Design Principles, Trusted Systems.					
Module:8 Contemporary Issues		2 hours			
		Total Lecture hours:		45 hours	
Text Book					
1.	Cryptography and Network Security-Principles and Practice, 8 th Edition, by Stallings				